

spaTransfer: transfer learning for single-cell and spatial transcriptomics data using non-negative matrix factorization

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Abstract

Recent advances in spatially-resolved transcriptomics have enabled profiling of gene expression in a spatial context, which has led to the generation of large-scale single-cell and spatial atlases with computationally-derived cell type or spatial domain labels. An increasingly important task with these data has become the transfer of cell type or spatial domain annotations from a given reference (or source) atlas into a new target tissue or sample. The reference and target datasets could be at different resolutions or measured on different experimental platforms. Here, we present a method to perform cross-platform transfer learning that takes as input single-cell or spatial domain labels from a reference atlas or dataset and transfers the labels to a target dataset at a similar or different resolution. Specifically, we use non-negative matrix factorization (NMF) on the reference data to identify factors associated with labels of interest and project these factors into the target dataset to label each new observation. We use a multinomial model with the factors as covariates and labels as the response to predict labels in the target dataset. In contrast to existing approaches, the advantage of our approach is interpretability, without compromising on accuracy. We demonstrate the performance of the method in two human brain tissues and show that our model identifies spatially coherent domains in the target datasets with concordance of marker gene expression. We implement spaTransfer in open-source software as an R package (github.com/cindyfang70/spaTransfer).

Keywords

Label transfer, non-negative matrix factorization, spatially-resolved transcriptomics, single-cell analysis, spatial domain detection, postmortem human brain

1 Introduction

The advent of spatially-resolved transcriptomics (SRT) has allowed for the analysis of gene expression at 2D resolution [1] leading to the generation of large-scale SRT atlases with labeled cell types or spatial domains [2–4]. As new SRT datasets are generated, an important task is to annotate the datasets [5, 6]. In single-cell/nucleus RNA-sequencing (sc/snRNA-seq) data, computational methods have been developed to transfer labels from a sc/snRNA-seq reference atlas to a target scRNA-seq [7–12] or scATAC-seq [13] dataset. These are broadly referred to as “single-cell reference mapping” or “label transfer” algorithms where the goal is to align the -omic (e.g. transcriptomic) profiles of samples in a target dataset to a reference atlas in order to transfer annotations at a single-cell resolution to the target dataset. In a recent perspective from Lotfollahi et al. [14], the authors described how the *interpretability* of single-cell reference mapping algorithms is important, especially in applications where the goal is to label complex populations of cells, for example, in response to perturbations. Furthermore, the authors suggest that computational methods that can decompose differential sources of variation and provide interpretability of the reference mapping algorithms will be beneficial to address future reference mapping challenges.

In application of SRT data, the goal of reference mapping is to transfer the labels (either cell types or spatial domains) in the reference atlas to a target dataset. Existing single-cell reference mapping algorithms such as singleR [15] and scmap [16] rely on measures of correlation between gene expression profiles of cells, but fail to leverage the spatial information. Further, imaging-based SRT technologies are limited in the number of measurable features and calculating the correlation between a target and a reference expression profile with different sets of features is challenging. Another challenge is the lack of an ideal molecular atlas due to rapidly emerging new SRT technologies that vary in their resolution. Current approaches to address this challenge include integrative methods such as Cell2location [17] or RCTD [18], which require users to provide paired sc/snRNA-seq reference samples as input. However, these approaches are better suited for decomposing the proportion of cell types rather than transferring cell type or spatial domain labels, as the sc/snRNA-seq reference does not contain any spatial information. Another broad category of integrative label transfer algorithms, such as Symphony [9], are algorithms that learn a low-dimensional representation of the reference dataset and use either cell or gene anchors to map the target dataset into the same low-dimensional space. Unfortunately, many of these algorithms rely on principal component analysis (PCA) for low-dimensional embeddings [14], which yields principal components (PCs) that can only be interpreted sequentially and not as standalone factors. Finally, recent methods have been developed to transfer labels from single-cell reference atlases to SRT data, but these approaches either are designed for similar sets of features or similar experimental platforms, reducing their impact as reference atlases grow across technologies or use deep learning [19–23], which limits their interpretability.

To improve compatibility with diverse SRT technologies across throughput and spatial resolution as well as interpretability to inform biological mechanisms, we developed a novel computational method to transfer labels (cell types or spatial domains) from a reference dataset to a target dataset using non-negative matrix factorization (NMF). This method offers unique advantages to address the challenges noted above by decomposing the observed gene expression into latent factors representing cell types or spatial domains, thus incorporating spatial information and circumventing direct comparisons of expression profiles from differing gene panels. As these factors can be associated with cell type or spatial domain labels, our approach provides direct interpretability of the annotations in the target dataset. In the next two sections, we provide a detailed overview of the method (spaTransfer) and describe the key innovations and advantages it has over existing approaches. Finally, we demonstrate the performance of spaTransfer on several real-world datasets and show how our approach offers clear interpretability without a loss in accuracy compared to existing methods.

2 Results

2.1 spaTransfer enables cross-platform transfer learning

spaTransfer is a reference mapping algorithm that transfers labeled spatial domains or cell types from a reference (or source) dataset to an unlabeled target dataset, where the datasets can be from different technologies, such as sn/scRNA-seq or SRT platform. Broadly, our approach begins by using non-negative matrix factorization (NMF) to identify factors (or patterns) in the reference dataset (**Figure 1**). The number of factors (rank), L chosen is either a user-specified number or is automatically determined [24]. NMF provides a way to factorize the gene expression data from single-cell or SRT experiments into two matrices with non-negative entries. Briefly, NMF decomposes a matrix $A \in \mathbb{R}_{\geq 0}^{G \times N}$, typically gene expression with G genes and N cells, into two matrices W and H , such that

$$A \approx WH$$

where $W \in \mathbb{R}_{\geq 0}^{G \times L}$ is a gene by rank matrix and $H \in \mathbb{R}_{\geq 0}^{L \times N}$ is a rank by cell matrix. For a given target dataset A' , we solve for H' in the equation $A' = WH'$, using the loadings matrix W from the reference dataset. In the setting where the reference dataset and target dataset differ in the observed genes, we subset W to only the genes shared between the two datasets. This yields a set of factors in the target dataset that correspond to the factors computed in the reference dataset, which represents the same biological processes, such as cell types or spatial domains. The reference factors (H') are then used as features in a multinomial model: regressing the reference factors to the spatial domain labels using the reference dataset. Leveraging the fitted multinomial model, the label for each observation can be predicted using the target dataset. Since it is possible for factors to represent platform-specific or sample-specific factors that are not desirable for downstream prediction, regularization is built into the multinomial model fitting process through elastic net regularization [25, 26].

2.2 Key innovations of spaTransfer

spaTransfer offers the following key innovations compared to existing approaches. First, spaTransfer enables the transfer of labels between datasets generated from different single-cell or SRT technologies with potentially different sets of features or at different resolutions. For example, spot-based data such as Visium can be thought of as “mini-bulk” data, with expression from multiple cells aggregated into each spot whereas imaging-based data such as Xenium is at single cell resolution. Despite the differing resolutions, there is evidence that NMF factors can address cross-platform challenges. For example, previous studies have shown that NMF factors learned on bulk RNA sequencing data can be transferred to scRNA-seq data, allowing for shared biological features that were first identified in the bulk data to be found in the single cell level data [27]. We take advantage of these properties to transfer information across different technologies. Specifically, we learn the latent factors that can be used to approximately generate the reference dataset through additive linear combinations. spaTransfer then estimates the composition of these latent factors in the target dataset. Based on the distribution of latent factors in each observation, spaTransfer is able to predict the annotation for the observation. Previous SRT studies have leveraged these latent factors for both discrete cell types and continuous patterns of gene expression [28, 29]. Overall, spaTransfer provides a unified framework for performing NMF on the reference dataset, transferring NMF factors to the target dataset, and leveraging these factors to fit predictive label transfer models.

Second, spaTransfer offers a computationally fast and biologically interpretable way to label cells or spatial domains in an unlabeled target dataset. Previously, NMF was challenging to use with high-dimensional transcriptomic datasets because of the computational burden to compute the factors. However, recent advances in optimization for matrix factorization for sparse matrices [24] have led to implementations of NMF that are comparable in computational speed to well-established algorithms that compute partial singular value decompositions (SVD) of large sparse or dense matrices [30, 31]. This greatly reduces the computational burden of using NMF in practice to find low-dimensional representations of transcriptomics data, such

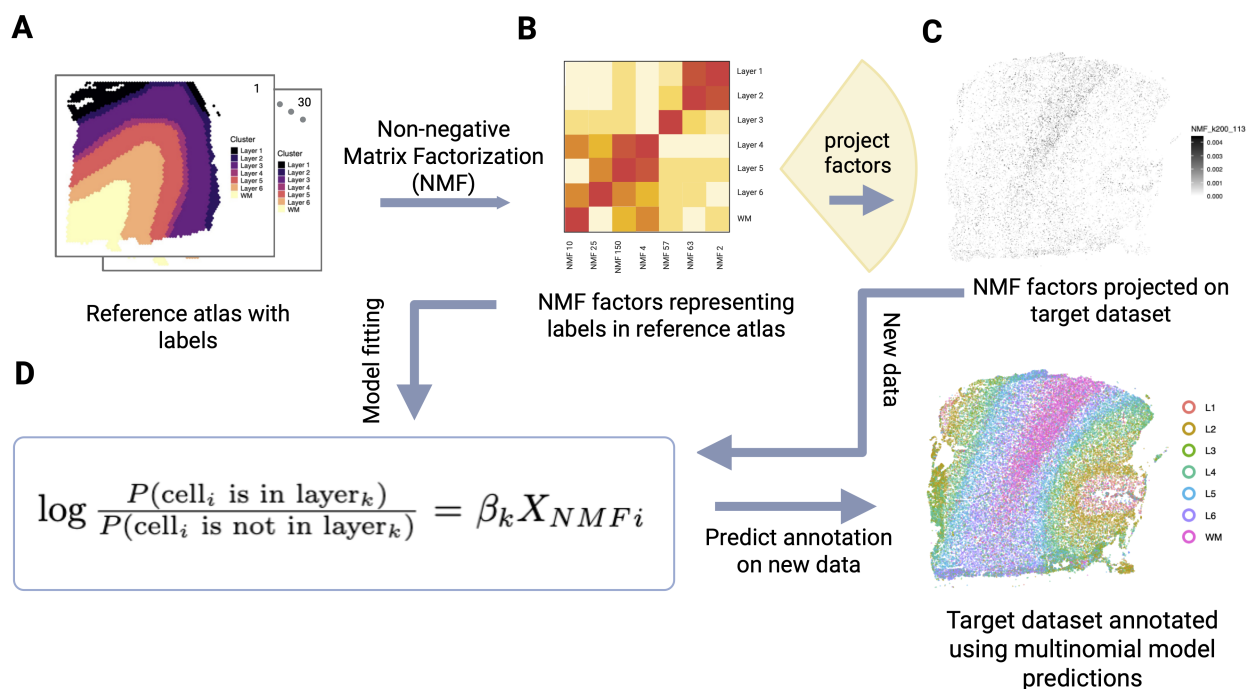


Figure 1: Overview of spaTransfer for transfer learning with single-cell and spatial transcriptomics data using non-negative matrix factorization. (A) Given a set of reference (or source) atlas with multiple samples and labeled cell types or spatial domains, the spaTransfer algorithm uses non-negative matrix factorization (NMF) on the source data to (B) identify factors associated with labels of interest. (C) Next, spaTransfer projects these factors onto the unlabeled target dataset. (D) To label each new observation (e.g. cell or spot), we use a multinomial model with the factors as covariates and labels as the response to predict labels in the target dataset. The primary advantage of our approach is interpretability without compromising accuracy of the label transfer. We demonstrate how our model identifies spatially coherent domains across multiple target datasets.

as for transfer learning. Furthermore, the additive nature of the algorithm enables us to sift out signatures representing interesting biological patterns, with each factor representing distinct biological processes[24]. The non-negative nature of NMF is particularly suited to transcriptomics data, as gene expression counts are inherently non-negative. This strategy allows for a more interpretable transfer learning algorithm than relying on PCA, which can result in negative values. NMF on the atlas yields a ‘loadings matrix’, which will be used to compute projections of the factors in the new dataset. These projected factors can then be used to identify tissue areas or cells with enrichment of biological patterns.

Third, spaTransfer can leverage information from multiple reference samples due to its usage of NMF. The additive nature of NMF allows for (i) each factor to be interpreted independently and (ii) removing factors that are highly correlated with batch effects or other unwanted technical artifacts, thus reducing the impact of potential erroneously labeled observations in the target dataset due to batch effects observed in the reference dataset.

Fourth, spaTransfer offers the option to smooth predicted labels in the target dataset across space. This step is especially helpful when transferring spatial domain labels as it increases the spatial contiguity of the predicted domains. Spatial smoothing of domains controlled by density of labels was also presented in BANKSY [32], but spaTransfer’s smoothing algorithm allows for domain-specific density parameters rather than a single global parameter.

2.3 spaTransfer provides interpretable factors to transfer labels from a reference atlas to a target dataset across experimental platforms with different resolutions

Next, we consider two examples where we apply spaTransfer and show how our model transfers spatial domain labels obtained in a reference dataset to an unlabeled target dataset at a single-cell resolution. The purpose is to demonstrate how spaTransfer is able to identify spatially coherent domains in the target dataset, which is profiled using a different experimental platform at a different resolution compared to the reference dataset.

2.3.1 Application to human dorsolateral prefrontal cortex

The granular cytoarchitecture of the human dorsolateral prefrontal cortex (dlPFC) is linked to the region's key role in higher-order cognitive functioning. Previous studies profiled human dlPFC tissue using snRNA-seq (single cell resolution) and the 10x Genomics Visium platform based on spots (multiple cells / spot resolution) [33–39]. The dlPFC contains six classic histological layers, which each have different distributions and densities of distinct cell types. To investigate the distribution of cell types within individual spatial domains (e.g. corresponding to histological layers), we generated single-cell resolution 10x Genomics Xenium data in postmortem human dlPFC tissue (6 tissue sections across 4 neurotypical brain donors) using the Human Brain Gene Expression panel (266 unique genes) (**Figure 2A, Table S1, Methods**). Next, we leveraged a reference dataset with labeled spatial domains measured on the 10x Genomics Visium platform with 30 tissue sections across 10 neurotypical brain donors [33] (**Figure 2B**).

Using the reference Visium dataset, we used spaTransfer to transfer the spatial domain labels to the unlabeled target dataset measured on the Xenium platform. First, we applied standard preprocessing and quality control (QC) considering negative control genes (**Figures S1-S3**) in the target dataset. Next, we identified factors using NMF in the reference dataset that were strongly correlated with the labeled spatial domains (**Figure 2C**) and other unwanted technical factors (**Figure S4**). For example, we found that factor 21 (NMF21) was associated with spots with high percentages of mitochondrial counts and factor 8 (NMF8) was highly correlated with the white matter (WM) domain in the reference dataset. This is an example of the interpretability provided by spaTransfer. Then, we visualized the factor loadings for NMF8 in 2D resolution on a Visium tissue section, which showed the factor weights were localized to the regions annotated as WM (**Figures 2D**). These factors were then projected into the Xenium dlPFC target datasets. The projected factors retained their spatial organization and aligned with known anatomy (**Figures 2E, S5**). Specifically, we found that spatial domain predictions from spaTransfer based on these transferred factors matched the expected laminar structure of the dlPFC, with individual cortical layers correctly annotated and sequentially positioned from the more superior meninges to the deeper WM layers (**Figures 2F, S6, S7**). Interestingly, some Xenium tissue slices such as those from donor Br8667 did not contain WM. This information was not provided to the method for label transfer, but spaTransfer was still able to accurately identify the structure of the tissue slice (**Figures S6, S7**).

2.3.2 Application to human dentate gyrus (DG)

Structural and functional synaptic plasticity in neurons of the hippocampus (HPC), particularly those within the dentate gyrus (DG) subfield, underlie several forms of learning and memory that change dynamically over the lifespan [41–43]. Many functions of the DG are topographically organized, and therefore, a better understanding of gene expression patterns across DG cell types, and how they change over development and aging, is important [40, 41, 44]. The human HPC and DG has been profiled using snRNA-seq across the lifespan, but this approach does not retain spatial information [45–49]. We recently used SRT to profile postmortem human DG tissue sections from 16 donors across the lifespan (N=5 infant, N=4 teen, N=4 adult, N=3 elderly) using the 10x Genomics Visium platform. Our study identified heterogeneity in spatial gene expression within the DG for molecular markers of proliferation, extracellular matrix, and neuroinflammation [40]. However, while the Visium platform preserves spatial context, information from individual cells is not available.

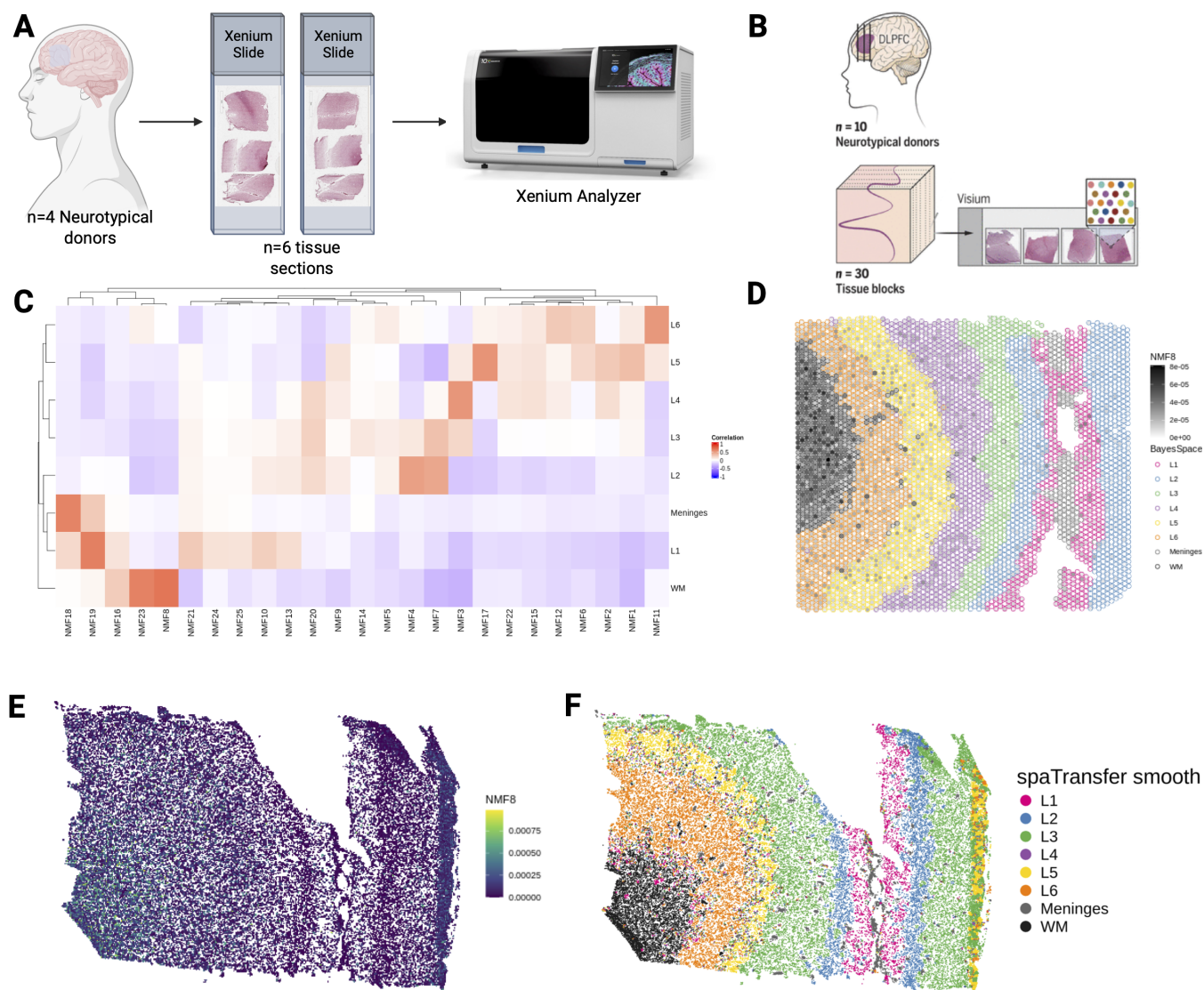


Figure 2: spaTransfer accurately transfers spatial domain annotations to a target dataset at single-cell resolution in human dorsolateral prefrontal cortex. We used two datasets both generated in postmortem human tissue from dorsolateral prefrontal cortex (dlPFC): (1) **(A)** an unlabeled target dataset at single-cell resolution measured on the 10x Genomics Xenium platform with 6 tissue sections generated across 4 neurotypical donors, and (2) **(B)** a reference dataset with labeled spatial domains measured on the 10x Genomics Visium platform with 30 tissue sections across 10 neurotypical donors [33]. **(C)** Heatmap of factors learned from the reference dataset using spaTransfer, which highly correlate with labeled spatial domains. **(D)** Spotplot of weights for factor 8 (NMF8) that is associated with white matter (WM) and visualized with labeled spatial domains in a tissue sample ('Br6471_Post'). **(E)** Cell-level plot of NMF8 projected in a Xenium tissue sample ('Br6471_Post_5434'). **(F)** Same tissue as **E**, but cell-level plot with colors representing the predicted spatial domains.

To facilitate better understanding of the human DG by utilizing both spatial and single-cell information, we generated SRT data on a subset of the donors from our previous study [40] using a precommercial prototype version of the 10x Genomics Xenium Prime 5K Human Pan Tissue and Pathways Assay. We profiled the postmortem human DG from four brain donors across the lifespan (N=1 infant, N=1 teen, N=1 adult,

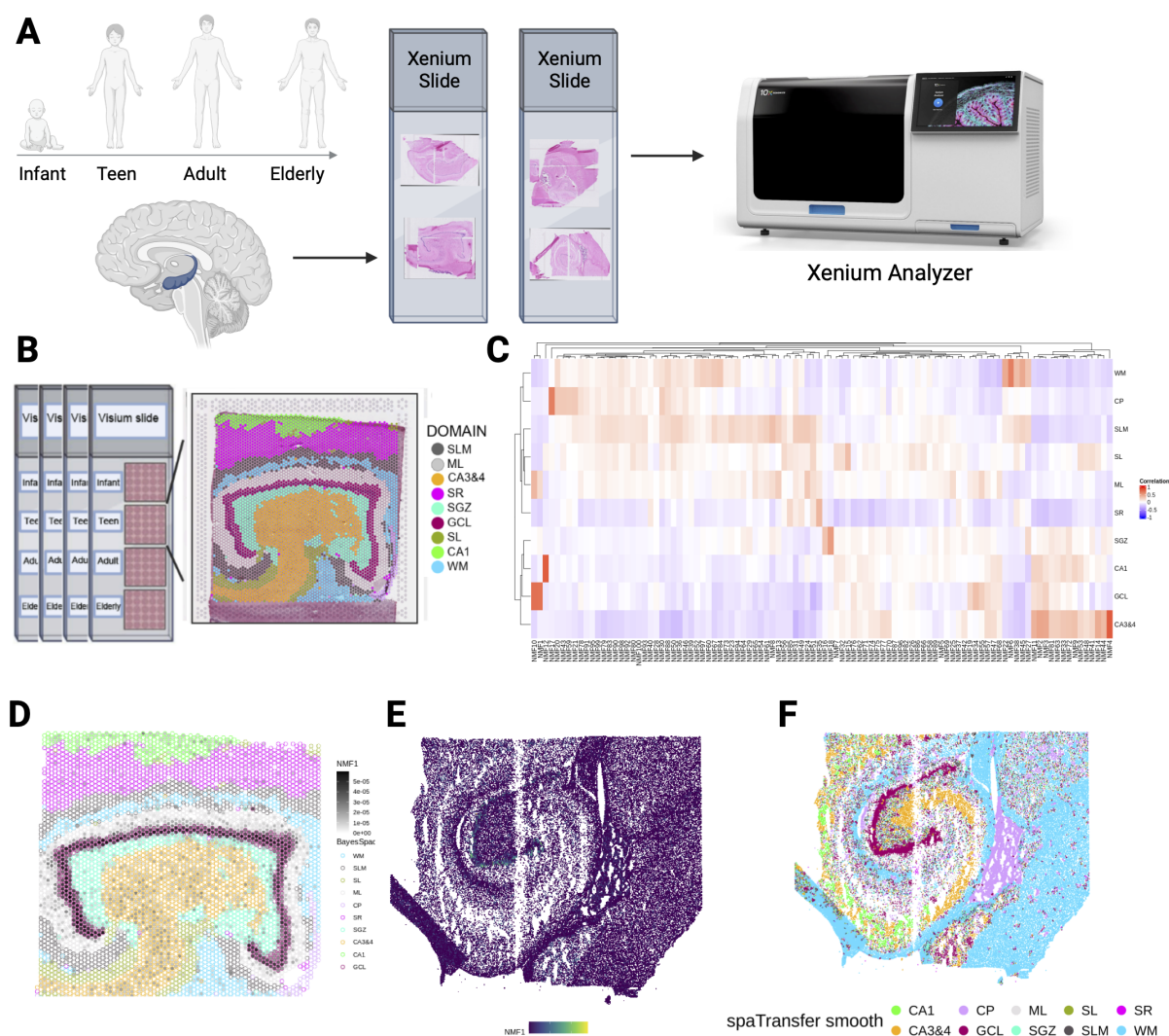


Figure 3: spaTransfer accurately transfers spatial domain annotations to a target dataset at single-cell resolution in postmortem human dentate gyrus (DG) of the hippocampus. We considered the following two datasets: **(A)** an unlabeled target dataset at a single-cell resolution measured on the 10x Genomics Xenium Prime platform with 4 tissue sections of human dentate gyrus from 4 neurotypical brain donors. Each donor represented one stage across the human lifespan (infant, teen, adult, elderly). **(B)** A reference dataset with labeled spatial domains measured on the 10x Genomics Visium platform, with 16 tissue sections from 16 donors, profiling the DG across the lifespan [40]. **(C)** Heatmap of factors learned from the reference dataset using spaTransfer, which highly correlate with labeled spatial domains. **(D)** Spotplot of weights for factor 1 (NMF1), which is associated with the granule cell layer (GCL) visualized with labeled spatial domains in tissue sample Br1412.teen **(E)** Cell-level plot of NMF1 projected in the Xenium tissue sample Br8533_infant. **(F)** Same as **(E)**, but cell-level plot with colors representing the predicted spatial domains.

N=1 elderly) (**Figure 3A**). Similar to the Xenium platform, Xenium Prime profiles gene expression at a single cell resolution, but allows for up to 5000 genes to be measured. First, we applied standard preprocessing and QC (**Figures S8-S10**). To annotate spatial domains in this newly generated dataset, we applied spaTransfer using the spatial domain labels from the reference atlas (N=16 Visium DG dataset) and the newly generated Xenium datasets as the target dataset. The Visium DG dataset was previously annotated by performing unsupervised, spatially-aware clustering with BayesSpace [50] followed by identifying canoni-

cal marker genes of each cluster [40]. These annotations consist of $k=10$ clusters each representing a unique spatial domain (**Figure 3B**). spaTransfer was able to identify latent factors representing spatial domains in the Visium reference dataset as quantified by Pearson correlation (**Figure 3C**). We also examined the spatial localization of these factors by plotting the factor values for each spot on the tissue. These plots showed that the spatial patterns of these factors are aligned with known anatomy (**Figure 3D**) and the BayesSpace annotations. Projecting these factors into the Xenium DG dataset using spaTransfer yielded factors that exhibited spatial localization concordant with anatomical structures. The spatial localizations of the projected factors were consistent between the reference and target datasets, for example NMF1 represents the granule cell layer (GCL) in the reference dataset (**Figure 3D**) and the projected NMF1 in the target datasets show spatial localization in the expected location of the GCL (**Figures 3E, S11**). Leveraging the NMF factors, which encode both spatial and gene expression information, spaTransfer was able to predict spatial domain annotations for each cell in the Xenium DG dataset. These predicted annotations show concordance with the known anatomy of the region and enable spatially-aware downstream analyses (**Figures 3F, S12, S13**).

Finally, we used the transferred labels in the Xenium DG data to explore cell-level differences (annotated with the spatial domains) across the lifespan. Specifically, we pseudobulked the cell-level counts across the gene for each spatial domain in each donor. After removing pseudobulk outliers related to the choroid plexus and other donor-specific artifacts (**Figure S14**), PCA was applied to the remaining 30 pseudobulk samples to identify the top components of variation in the data. We found that the first two principal components (PCs) (26% and 21% of variance explained, respectively) appear to be driven by differences between broad cell classes (neurons, white matter, neuropil) (**Figure S15**). Separation of infant samples from the other age groups is seen along PC3 (15% of variance explained), which matches findings in the original paper [40]. This suggests that gene expression patterns in the DG during early life significantly differ from that in later life (**Figure S16**).

2.4 spaTransfer accurately annotates spatial domains with high spatial contiguity and cluster validity

To evaluate the performance of our method, we considered two sets of evaluations. First, we aimed to evaluate the accuracy of annotating spatial domains using spaTransfer compared to two widely used single-cell reference mapping algorithms, namely singleR [15] and Seurat [10, 51]. Our rationale was to select algorithms that were designed to be agnostic to leveraging spatial information in the reference atlas, but we wanted to evaluate if they could accurately label spatial domains at single cell resolution, if provided a reference dataset with those labels. Second, we considered the performance of spaTransfer compared to unsupervised spatial domain detection algorithms to evaluate the spatial contiguity and cluster validity. Our rationale for this was to evaluate the performance of algorithms that transferred labels from a reference dataset compared to algorithms that perform *de novo* identification of spatial domains in a new dataset using unsupervised clustering algorithms. We recognize these can be complementary approaches, but with the increasing availability of reference datasets and atlases, we aimed to answer this question more formally.

Towards the first goal, we considered the human dlPFC data profiled on the 10x Genomics Visium platform [33] as described in Section 2.3.1. Specifically, we split the 30 tissue sections from this dataset in half where 15 were used as the reference dataset and the other 15 were used as the target samples. We used the originally reported spatial domains as the labels in the reference atlas for spaTransfer, singleR [15], and Seurat [10, 51] and transferred the labels using these algorithms (see **Methods** for details). To compare the accuracy between these methods using the 15 target samples, we used the originally reported data-driven spatial domain labels (using BayesSpace [50]) in the 15 target samples to compare against the transferred labels from the three algorithms (**Figure 4A**). To assess the accuracy of the label transfer methods, we computed the adjusted Rand index (ARI) and normalized mutual information (NMI) (**Figure 4B**). We found that spaTransfer with spatial smoothing (median ARI: 0.45, median NMI: 0.52) outperforms both singleR (median ARI: 0.16, median NMI: 0.26) and Seurat (median ARI: 0.43, median NMI: 0.50) in terms of both ARI and NMI, as higher values indicate better performance.

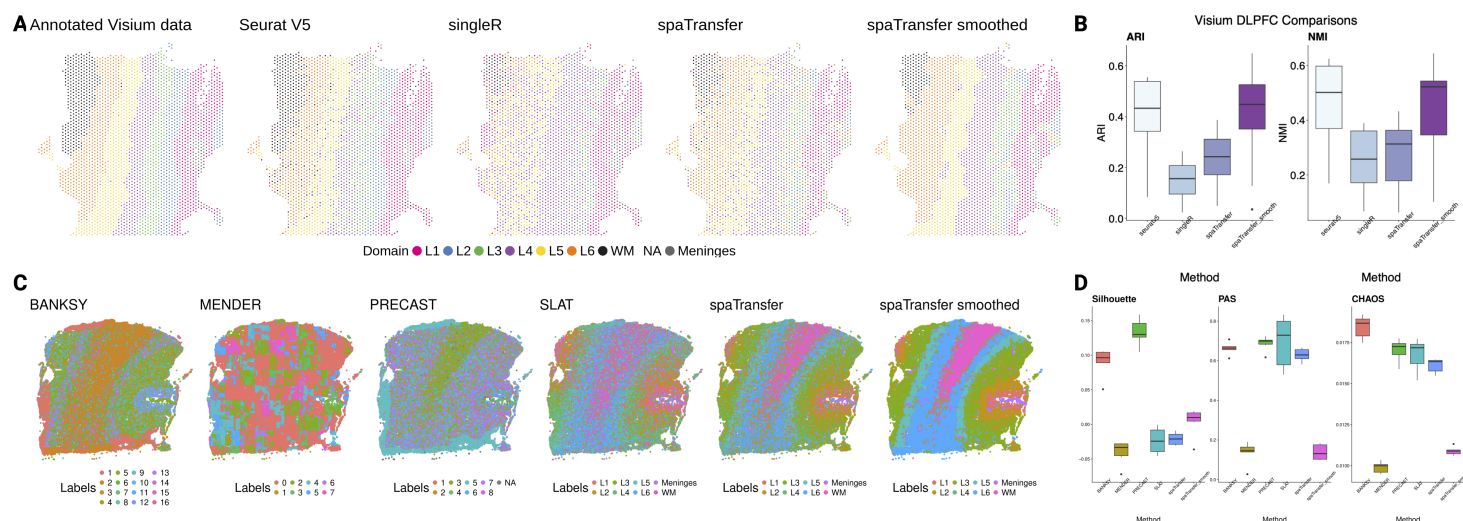


Figure 4: spaTransfer accurately annotates spatial domains with high spatial contiguity and cluster validity. (A) Predictions from all transfer-learning based methods on Visium tissue Br6471_ant. (B) Boxplots of ARI and NMI for each of the different label transfer methods, computed across the target set Visium samples. spaTransfer shows better average performance than singleR in terms of both ARI and NMI. spaTransfer also shows comparable performance to Seurat. (C) Cluster labels from all unsupervised methods on Xenium tissue Br2743_Mid_5548. (D) Silhouette index, proportion of abnormal spots (PAS), and CHAOS metrics computed for all unsupervised methods across all 6 Xenium dlPFC samples. spaTransfer has the best performance in terms of the PAS and CHAOS metrics.

We also evaluated the performance of spaTransfer on Xenium data against widely used spatial domain detection methods that can be applied to imaging-based SRT, namely BANKSY [32], MENDER [52], PRECAST [53], and SLAT [54]. For this evaluation, we continued with the same human dlPFC data profiled on the 10x Genomics Xenium platform as described in Section 2.3.1. To evaluate the biological validity of the predicted domains, we used an internal validity metric that did not depend on a ground truth label, namely the silhouette value [55], which was calculated on the top 50 PCs of normalized gene expression. The maximum silhouette value is 1 and the minimum value is -1, with higher scores indicating better performance. Next, we aimed to evaluate the spatial contiguity of the predicted domains or how “domain-like” each cluster was, with the assumption being that spatial domains should exhibit contiguity in physical space. Towards this goal, we considered two metrics: (i) proportion of abnormal spots (PAS) [56] and (ii) spatial chaos (CHAOS) [56]. For both of these metrics, lower values of the metrics indicate more spatial contiguity. In our evaluations, we found that spaTransfer reported the highest spatial contiguity compared to reference-free methods as quantified by PAS, and is competitive in terms of CHAOS (Figures 4D, S17). Notably, the performance of spaTransfer was lower using the silhouette index in our evaluations. However, the silhouette index was computed using principal components of gene expression, meaning it was inherently biased towards BANKSY, MENDER, and PRECAST, which all rely on PCA for dimension reduction before clustering.

3 Discussion

We introduced a new computational method (spaTransfer) to transfer the labels (cell types or spatial domains) from a reference atlas to an unlabeled target dataset that prioritizes interpretability and can be used across different platforms with different sets of features. The core of our method using non-negative matrix factorization to learn a set of latent factors in the reference atlas and then use a multinomial model to predict the labels (e.g. spatial domains or cell types) using the latent gene expression patterns.

We demonstrate the performance of spaTransfer using two reference atlases generated using the Visium

platform [33, 40] to transfer the labels to two newly generated Xenium datasets (human dlPFC and human dentate gyrus). These datasets complement existing Visium datasets by providing higher-resolution spatial measurements, enabling the study of transcriptional heterogeneity at cellular resolution. Most notably, we demonstrated the ability of spaTransfer to capture and transfer gene expression patterns across different platforms with different sets of features, while also balancing biological validity and spatial contiguity as evaluation metrics. In terms of performance, spaTransfer accurately identified spatial domains in these datasets and also produced more spatially contiguous domains, which aligned with expected neuroanatomy.

Towards limitations, we recognize that several deep learning algorithms have already been developed to transfer labels between single-cell reference atlases and spatial target datasets. However, these tools lack transparency and interpretability, which are necessary if there is increased reliance on reference atlases to transfer cell type or spatial domain labels to newly generated target datasets. We also acknowledge that NMF requires a user-defined parameter (L or the number of latent factors), which could impact the performance of the label transfer algorithm if not chosen carefully. This is an important area of future research, which has become more computationally feasible due to advances in sparse matrix factorization algorithms [24]. Finally, we recognize that transferring labels from transcriptome-wide technologies to targeted gene panels will be inherently limited by the gene panel. Although our simulation analyses demonstrated that subsetting a dataset to 25% of its original features preserves the distribution of NMF factors **Figure S18**, results will likely depend on how informative each gene or feature is. Thus, it is important to carefully select genes during panel design to consider whether genes that are relevant for label transfer will be present.

4 Methods

4.1 Datasets

4.1.1 Publicly available datasets using 10x Genomics Visium

human dlPFC. Gene expression counts were obtained from the *spatialLIBD* R package [57] as described in Huuki-Myers et al. [33]. This reference dataset contains 3 tissue regions spanning the rostral-caudal axis of the dlPFC from each donor, with a total of 10 neurotypical adult donors. The 10x Genomics Visium Spatial Gene Expression platform was used to profiling the gene expression of postmortem dorsolateral prefrontal cortex (dlPFC) in human brain across $N=30$ tissue sections, 28,916 genes, and $n=113,927$ spots. The reference data was previously annotated using the spatially-aware data-driven clustering method from BayesSpace [50] and spatially registering BayesSpace clusters to manual spatial domain annotations. These spatial domain annotations consist of 7 different labels: Layers 1 through 6 (L1-6) and white matter (WM).

human DG. Gene expression counts were obtained from Zenodo [58] as described in Ramnauth et al. [40]. This reference dataset consists of $N=16$ tissue samples of anterior hippocampus from fresh-frozen postmortem human brains. The dentate gyrus (DG) was dissected from these slabs and mounted on Visium slides, with each capture area containing a unique donor. This resulted in 30,217 genes measured across $n=68,685$ spots. The reference dataset clustered using BayesSpace [50] for a total of $k=10$ clusters which were then annotated with canonical hippocampal spatial domains based on expression of marker genes and comparison with histology from H&E staining.

4.1.2 Postmortem human tissue samples

Postmortem human tissue samples. Postmortem human brain tissue from donors of European ancestry were obtained at the time of autopsy following audiotaped witnessed informed consent from legal next-of-kin, through the Maryland Department of Health IRB protocol #12-24, and from the Western Michigan University Homer Stryker MD School of Medicine Department of Pathology, and the Department of Pathology at University of North Dakota School of Medicine and Health Sciences, all under the WCG protocol #20111080. One additional sample was obtained by material transfer agreement from the NIMH Brain Collection Core,

under protocol #90-M-0142. Curation, tissue handling, processing, and quality control measures have been detailed previously by Lipska et al. [59]. This study included four HPC samples and four dlPFC samples, and demographic information for all donors is listed in **Supplemental Table S1**. A small piece of dlPFC or DG from each donor was dissected under the visual guidance of a neuroanatomist with a handheld drill on dry ice. Microdissected dlPFC and DG tissues were stored at -80C until cryosectioning, and anatomical orientation was validated as previously described by Huuki-Myers et al. [33], Ramnauth et al. [40].

4.1.3 Generation of human dlPFC using 10x Genomics Xenium

Tissue processing. The dlPFC data shown in this study were collected from 4 adult neurotypical donors (**Table S1**). Cryosections from dlPFC tissue samples were mounted onto Xenium slides with 3 tissue samples from 3 donors on each slide. We collected adjacent replicates across the 2 Xenium slides for 2 donors. Sectioning specifications for dlPFC samples are listed in the 10x Genomics protocol CG000579 (Rev C).

Xenium In Situ Workflow. Fixation and permeabilization steps were performed as described in protocol CG000581 (Rev C, 10x Genomics). Probe hybridization, ligation, amplification, autofluorescence quenching, and nuclei staining was performed as specified in CG000581 (Rev C, 10x Genomics) using the 266 gene human brain panel (Xenium Human Brain Gene Expression Panel, Part No.1000599). All slides were loaded onto the Xenium analyzer at the Single Cell -Transcriptomics Core (SC-TC) at the Johns Hopkins University School of Medicine and regions of interest (ROIs) were selected according to the 10x protocol CG000584 (Rev D).

4.1.4 Generation of human DG using 10x Genomics Xenium prime and multimodal segmentation

Tissue processing. The DG data shown in this study came from four donors of differing ages (N=1 infant, N=1 teen, N=1 adult, N=1 elderly) (**Table S1**). Cryosections from fresh frozen DG tissue samples were mounted onto Xenium slides with two tissue samples from different donors on each slide. The infant and adult samples were mounted onto three slides and the teen and elderly were mounted onto another three slides. Sectioning specifications for these samples are listed in the 10x Genomics protocol CG000579 (Rev C). Slides with frozen tissue sections were shipped on dry ice to 10x Genomics, where the samples were processed in two separate batches.

Xenium In Situ Workflow. To perform Xenium in situ analysis (10x Genomics) on DG samples, fixation and permeabilization steps were performed as described in protocol CG000581 (Rev C, 10x Genomics). Probe hybridization, ligation, amplification, autofluorescence quenching, and multimodal cellular segmentation staining was performed by 10x Genomics. The gene panels from both batches contain a 4200 gene base panel and a choice between two high expression panels of 289 genes. Genes in the two high expression panels were compared to differentially expressed genes from an existing Visium study of the hippocampus across the lifespan. The panel with more overlap with genes that were differentially expressed in age groups was selected. Unlike the first batch, we were also able to provide custom gene targets in the second batch. This custom panel consisted of 15 genes that are important for expression profiling in the DG. These genes are *SLC17A7*, *SLC17A6*, *CD74*, *DCX*, *PROX1*, *LAMP5*, *TTR*, *FIBCD1*, *NECAB1*, *FN1*, *PPFIA2*, *MPPED1*, *SLC12A5*, *METTL7B*, and *SEMA5A*.

4.2 Analysis of Xenium datasets

4.2.1 Xenium Onboard Analysis

After imaging on the Xenium Analyzer instrument, the dlPFC and DG data were processed using the 10x Genomics Xenium On Board Analysis (XOA) pipeline. The XOA pipeline processes the volumes generated across different fields of view (FOVs), fluorescence channels, and probe hybridization and imaging cycles. The dlPFC datasets were processed using XOA version 1.5 and the DG datasets were processed using a prototype version of XOA 3.0.0.0. In both versions of the XOA pipeline, image processing and RNA decoding is performed.

4.2.2 Quality control and preprocessing

Outputs from the XOA pipeline were used as input for quality control and preprocessing of the Xenium data prior to performing downstream analyses. Specifically, the `cell_feature_matrix.h5`, `cells.csv.gz`, `cell_boundaries.csv.gz`, and `nucleus_boundaries.csv.gz` files were read into R as using the *SpatialExperiment* package [60]. The `cell_feature_matrix.h5` file contains high-quality transcript counts that are found within cells. High-quality transcripts were identified by the XOA pipeline based on a Phred-style score, which is then calibrated using control probes and codewords. Several types of control probes and codewords are used in the Xenium workflow, and a brief description of each type of negative control is available from 10x Genomics. We refer to these collectively as ‘negative controls’. Only counts with a calibrated score of ≥ 20 are deemed high-quality and retained for downstream analyses. The cell feature matrix is then used as the counts matrix in a *SpatialExperiment* object. The `cells.csv.gz` file contains the spatial coordinates of each cell as well as each cell’s area, its nucleus area, the total number of counts in transcripts, the number of counts from negative controls, and the total number of overall counts. Due to the limited gene panel in Xenium, we do not quantify reads mapping to the mitochondrial genome and therefore do not use this as a QC metric. Instead, the *scater* R package [61] was used to identify and remove low quality cells based on thresholding counts from negative controls. Cells that were 5 or more median absolute deviations (MADs) higher than the median value of any negative control were deemed to be low quality and removed from downstream analyses. Additional filtering was performed to remove cells with zero total counts and a nucleus area of less than or equal to 200 microns squared, based on expected cell sizes in the human brain. For the Xenium dlPFC dataset, 17,972 cells were discarded out of a total of 318,063 cells across six tissue sections (**Figures S1-S3**). For the Xenium DG dataset, this resulted in 25,387 cells discarded from the total of 532,331 cells across four tissue sections (**Figures S8-S10**).

Feature selection typically follows after normalization in scRNA-seq analyses to facilitate computation due to the large number of genes. However, the Xenium datasets only contain a few hundred to a few thousand genes, allowing this step to be bypassed and all gene features to be used in downstream analyses. Features representing negative controls were excluded from downstream analyses as they do not provide additional biological information.

4.2.3 Spatial domain annotation of Xenium dlPFC data

The human Visium dlPFC dataset [33] was used as the reference for label transfer with *spaTransfer* to the Xenium dlPFC data as the target. A rank of 25 was used for the number of NMF factors. Spatially-aware label smoothing was applied using a 10 nearest neighbors graph.

4.2.4 Spatial domain annotation of Xenium DG data

The human Visium DG dataset [40] was used as the reference for label transfer with *spaTransfer* to the Xenium DG data as the target. A rank of 100 was specified for the number of NMF factors chosen. Spatially-aware label smoothing was then run on the transferred labels in order to obtain more contiguous domains and reduce noise in the Xenium annotations. For each sample, a 15 nearest neighbors graph

was constructed based on spatial coordinates and label smoothing was performed across this graph (see **Methods**).

4.3 Overview of spaTransfer

4.3.1 Motivation for using non-negative matrix factorization

The primary motivation behind using NMF in spaTransfer was its ability to decompose a dataset into its constituent parts, which are interpretable as standalone factors, allowing for the identification of signal that is conserved between different experimental platforms. For example, a group may have previously generated Visium data for their tissue of interest and now want to use their Visium dataset to annotate a newly generated Xenium dataset. The observational unit in the Visium platform known as a “spot” is a circular region spanning 50 microns in diameter. Due to its size, each spot typically contains multiple cells and users are only able to observe the transcriptional profile of the entire spot rather than the individual cells within. In contrast, Xenium is an imaging-based platform that offers sub-cellular resolution measurements of gene expression, and observations can be aggregated to the single cell level. To annotate a single-cell resolution dataset with the spatial domain labels from a spot resolution dataset, spaTransfer identifies latent factors in the reference atlas that represent biological processes of interest, in this case spatial domains. The same set of latent factors are then computed in the unlabeled target dataset, but at a single-cell resolution. Previous work has shown that this transfer of latent factors across datasets of different resolutions accurately preserves spatially organized biological signals [28, 29]. Similar to previous work, we found here that spaTransfer is able to leverage the preserved information encoded by the latent factors to transfer labels between datasets of different features and resolutions.

4.3.2 Using NMF to identify latent factors in reference atlas

NMF was originally proposed by Lee and Seung [62] as an alternative to PCA and vector quantization. The original paper found that NMF applied to images of faces resulted in a decomposition into factors representing individual facial landmarks such as the nose, mouth, or eyes. Similar to PCA, NMF factorizes a matrix into two lower-rank non-negative matrices W and H , but with the additional constraint of W and H being matrices with non-negative entries. Additionally, the factorization can instead solve for W and H in the following equation:

$$A \approx WDH \quad (1)$$

Where D is a scaling factor that ensures each NMF factor sums to 1. spaTransfer uses the *singlet* R package [63] which relies on the NMF implementation in RcppML [24]. If A is a gene expression matrix with log-normalized gene expression counts from single-cell or SRT experiments with each row representing a gene and each column representing a cell, then W is a gene by rank matrix and H is a rank by cell matrix. Similar to PCA, W can be thought of as a loadings matrix and H is a factors matrix, with each row in H representing a factor. The non-negativity constraint on the factorization yields factors that can be interpreted additively and independently of other factors, unlike in PCA [62]. Thus, when an appropriate rank is used for the factorization, NMF factors can be interpreted as representing a particular spatial domain or cell type.

Briefly, the NMF model is fit using alternating least squares. First, W is randomly initialized and a least squares procedure is used to find an estimate for H . Then, least squares is used again with H held constant to find a better estimate for W . This process is repeated until the relative change of and between iterations is smaller than the user-specified tolerance parameter.

When performing NMF on a reference dataset, spaTransfer offers two options for specifying the rank of the factorization L . The first option is for the user to provide L based on domain knowledge of the tissue. For the purposes of label transfer, we recommend that at minimum L should be equal to the number of levels in the annotation. This provides a necessary, but not sufficient condition for each level of the annotation to be uniquely represented by a NMF factor. However, it has been previously observed that NMF factors

can also represent technical artifacts [24, 27–29] and thus a higher value of L may be needed to ensure representation of all annotation levels. Conversely, a value of L that is too high can result in overfitting and poor prediction performance. The second option presented by spaTransfer aims to address this issue. If the user does not specify a value for L , then by default spaTransfer will run automatic rank determination using the `RunNMF()` function from the `singlet` package [63]. The automatic rank determination approach uses cross validation to select an optimal value for L while accounting for overfitting. Specifically, the automatic rank determination function will “mask” a user-specified percentage of the entries in the matrix A . By default, this is set to 5% of the total number of entries in A . These values are masked by replacing them with random normal variables. An NMF model is then fit at low value of L . The performance of the NMF model at this low rank is then evaluated using the mean squared error between the entries that were previously masked and the model’s predicted values for those entries. This process is repeated for different values of L until a value that minimizes the mean squared error is found. By default, the automatic rank determination function runs three independent replicates of this procedure to assess consistency of results across different initializations.

4.3.3 Transfer of NMF factors to target datasets

Once factors are extracted from the reference dataset (A), the gene by rank loadings matrix (W) is used to predict NMF factors on the expression values of the target dataset (A'). The prediction can be expressed as solving for H' in the following equation:

$$A' = WH'$$

where A' is the log-normalized counts matrix of the target dataset and W is the loadings matrix fitted on the reference dataset. Then, the least-squares estimate of can be computed as follows by solving the following equation

$$WW^T h'_j = W^T A_j$$

for each h'_j , where h'_j refers to the j th column of H . This was done using the `project` function from `RcppML` [24] following previously established workflows [28, 29].

In the case of cross-platform label transfer, the number of genes often differs between the reference and the target datasets, making matrix multiplication impossible due to incompatible matrix dimensions. To address this issue, the loadings matrix is subsetting after fitting NMF to contain only the genes that are found in both the reference and target datasets. The subsetting loadings matrix is then used in the `project` function from `RcppML` [24] to predict factors in the target dataset. In the case of the Xenium DG label transfer with the Visium DG dataset as the reference, we subset the original full-transcriptome loadings matrix to only the 5000 genes that are in the Xenium data. To ensure that the predicted factors are still meaningful, we simulated fitting NMF on randomly generated matrices and reconstructing factors using a subsetting loadings matrix. These simulations showed that subsetting to 10% of the original features will preserve the distribution of the majority of the factors, and subsetting to 25% of the original features preserves the distribution of all factors (**Figure S18**), assuming that all features are equally informative. Thus, retaining 5000 out of 20,000 genes should be sufficient for predicting factors in the target dataset.

4.3.4 Multinomial annotation prediction model

spaTransfer uses a multinomial model to generate annotation predictions on the target dataset with the annotation as the response as the selected NMF factors as the covariates. The multinomial model is implemented through the `glmnet` R package [26]. Let C denote the cell type or spatial domain annotations in the reference dataset. For a given observation in class c in C , we model

$$\Pr(C = c | X = x) = \frac{e^{\beta_{0c} + \beta_c^T x}}{\sum_{\ell=1}^C e^{\beta_{0\ell} + \beta_\ell^T x}}$$

where x denotes the NMF factors for the observation. This yields a linear predictor fit for each class of the annotations. Elastic net regularization [25] implemented in the *glmnet* R package [26] is used to prevent overfitting, resulting in the following log-likelihood:

$$\ell(\{\beta_{0c}, \beta_c\}_1^C) = -\left[\frac{1}{N} \sum_{i=1}^N \left(\sum_{c=1}^C y_{il}(\beta_{0c} + x_i^T \beta_c) - \log \left(\sum_{\ell=1}^C e^{\beta_{0\ell} + x_i^T \beta_{\ell}} \right) \right)\right] + \lambda \left[(1 - \alpha) \|\beta\|_F^2 / 2 + \alpha \sum_{j=1}^p \|\beta_j\|_q \right]$$

Where y_{il} is an indicator denoting whether observation i is a member of class l , α is the elastic net parameter, controlling the weighting on the lasso and ridge penalties, and λ controls the strength of the penalty term. The parameter q controls the type of lasso penalty used. If q is set to 1, each of the $K \times p$ coefficients each have their own lasso penalty, and can be zero or non-zero independent of other coefficients. If q is set to 2, the grouped lasso penalty is used and the K coefficients corresponding to the same variable will all be either zero or non-zero. We chose to use the grouped lasso penalty to ensure the same set of factors were selected across each of the classes. Further details of the model-fitting procedure can be found in *glmnet* [26].

4.3.5 Label smoothing

After annotations have been transferred into the target datasets, *spaTransfer* provides an optional spatially-aware smoothing algorithm to increase coherency of transferred domains. The foundation of the smoothing algorithm follows a previously established method [32] which builds a k -nearest neighbors (kNN) graph based on the spatial coordinates for each cell using the *dbscan* R package [64]. The number of neighbors is specified by the user. Then, for each cell, the proportion of neighbors labeled with each level in the annotations is computed. If any of the level proportions in the neighborhood is greater than the user-specified threshold, the cell is then assigned to the level with the greatest proportion. Otherwise, the cell retains its original label. This process is repeated for a user-specified number of iterations, with the number of cells being reassigned generally decreasing with each iteration. Our modification to this algorithm allows for a different threshold to be used for each level in the annotations, rather than having the same threshold for all levels. This is motivated by the different cell densities exhibited by spatial domains.

4.4 Downstream analyses of spatial domains

4.4.1 Pseudobulk processing of DG data

After annotating each cell's spatial domain identity using *spaTransfer*, we sought to explore the differences between spatial domains across the lifespan. To this end, we summed up the raw counts for each gene across all cells of each spatial domain using the `aggregateAcrossCells()` function from the *scuttle* R package [61]. This resulted in 40 pseudobulked samples (4 donors x 10 spatial domains) with each one representing a unique donor-spatial domain combination. The choroid plexus (CP) pseudobulk samples were removed as the CP is a secretory tissue that has been shown to account for the majority of variation when included in PCA [40]. The pseudobulk samples representing the stratum radiatum (SR) domain were also removed as we found that they also drove PC1 (57% of variance explained) (**Figure S14**). We found a similar result with the pseudobulk samples representing the CA1 and SL regions in the teen donor. This is likely due to the CA1 region being absent in the tissue section and thus any cells with the CA1 label are misclassified. Because only a small number of cells had these labels (197 CA1 cells, 168 SL cells), the teen CA1 and SL pseudobulk samples were also excluded from downstream analyses. This procedure was done separately for each donor and the pseudobulked samples were combined into one large `SpatialExperiment` [60] object, with each pseudobulk sample (column) representing a unique donor-domain combination. Genes that are lowly expressed in the spatial domains are removed from the pseudobulked samples using `filterByExpr()` in the *edgeR* R package [65]. Finally, log-normalized counts were calculated from the pseudobulk samples using `calcNormFactors()` in *edgeR*. PCA was used to identify the primary sources of variation in the pseudobulked dataset using the log-normalized counts computed above using `runPCA` from *scater* [61] using 10 PCs.

4.5 Comparison against existing methods

4.5.1 Comparing accuracy of predicted labels

To evaluate the performance of spaTransfer, we ran three existing single-cell reference mapping algorithms. Specifically, these were singleR [15], Seurat (v5) [10, 51]. We benchmarked the performance of these methods on the Visium dIPFC dataset [33]. Half the samples were used as the reference atlas and the remaining half were used as the target samples. Each of the methods were then used to predict spatial domain annotations on the target samples. To quantify the performance of each method, the adjusted Rand index (ARI) [66, 67] and normalized mutual information (NMI) [67] were computed between the labels annotated by the original authors of the manuscript in the reference and the predicted labels from spaTransfer or the other two algorithms (singleR and Seurat).

4.5.2 Comparing spatial contiguity and internal validity of predicted annotations

Due to the lack of ground truth labels for the Xenium datasets, we were unable to compute external evaluation metrics to evaluate the accuracy of transferred labels. Thus, we compared the performance of spaTransfer to BANKSY [32], MENDER [52], PRECAST [53], and SLAT [54] by computing metrics quantifying the spatial contiguity and internal validity metrics of transferred spatial domain labels (described below). All of these methods perform unsupervised and reference-free spatial domain detection, with the exception of SLAT. SLAT is a reference mapping method that uses a Visium dataset as the reference, and transfers labels from the Visium dataset to a target Xenium dataset. However, since it is only designed for the Visium to Xenium setting, we chose to benchmark it along with the other unsupervised methods due to the lack of Xenium ground truth labels.

For each of the methods, we computed the proportion of abnormal spots (PAS) [56] and the CHAOS metric [56] to evaluate the spatial contiguity of the predicted annotations. The PAS is defined as the number of abnormal cells out of the total number of cells in a tissue section. An abnormal cell is defined as a cell that has a different spatial domain label that is different from 6 or more of its nearest 10 neighbors [56]. The CHAOS metric is computed for each tissue section by first creating a 1-nearest neighbor graph for each spatial domain using the spatial locations of each cell. Then, for each pair of cells in domain r ,

$$E_{r,i,j} = \begin{cases} e_{i,j} & \text{if cell } i \text{ and cell } j \text{ are neighbours} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

and

$$CHAOS = \frac{\sum_{r=1}^R \sum_{i,j} e_{i,j}}{N}$$

where R is the total number of spatial domains and N is the total number of cells in the tissue section [56]. Finally, the Silhouette index was used to evaluate the internal validity of clusters. For each tissue section, PCA was first computed on the log-normalized counts using the *scanpy* Python package [68] with default parameters. The Silhouette score was then computed using the *scikit-learn* package [67], with euclidean distance as the metric.

4.5.3 Inputs and settings for the methods

We applied the two existing label transfer methods to our Xenium dIPFC dataset. For singleR and Seurat, we supplied the Visium dIPFC dataset as the reference. For singleR and Seurat, we followed the author-recommended preprocessing workflows provided in each method’s tutorials. For spaTransfer, we used the log2-normalized gene expression and a factorization rank of 25. Batch correction was not performed prior to running any of the methods. We used the same set of reference and query datasets for all three methods, thus allowing for a fair comparison between methods. For all three methods, we performed quality control and pre-processing of our datasets as described above and then followed the author-provided tutorials and their recommended parameter settings.

We also applied four domain detection methods that are applicable to imaging based SRT to our data. For BANKSY [32], MENDER [52], PRECAST [53], and SLAT [54], we provided the Xenium dIPFC dataset as the input for each method, and followed the authors' tutorials to preprocess the data and select parameters.

4.6 Software and data availability

spaTransfer is freely available as an R package on GitHub (github.com/cindyfang70/spaTransfer). Code to reproduce all preprocessing, analyses, and figures in this manuscript is available from GitHub at github.com/LieberInstitute/spatransfer-manuscript. We used spaTransfer version 0.0.0.9000 for the analyses in this manuscript. Analyses were performed with R version 4.3. Data visualizations were performed with *ggplot2* [69] and *escheR* [70] R packages. The processed data associated with the analyses in this manuscript are available at zenodo.org/records/17872293. The raw Xenium data files are currently being uploaded to GEO and will be available in a future update.

4.7 Acknowledgments, Funding, Authorship Contributions

Acknowledgments. Portions of some figures were created with BioRender.com. We thank 10x Genomics for reagents and early access to a precommercial version of the 5000-Plex Xenium Prime assay. The authors extend their gratitude to donor families who contributed tissue and clinical information for this study. We would also like to thank Amy Deep-Soboslay and James Tooke at the Lieber Institute for Brain Development for their exceptional efforts in the curation of the brain samples for this investigation. We thank the maintainers of the Joint High Performance Computing Exchange (JHPCE) compute cluster at Johns Hopkins Bloomberg School of Public Health for providing essential computing resources. We also thank Hicks lab for their constructive comments and suggestions. We thank Yi Wang for suggesting the idea of a multinomial model to predict labels.

Funding. This project was supported by NIH/NIGMS R35GM150671 (SCH) to support the computational methods development. Xenium data generation was supported by NIH/NIMH R01MH123183 (KRM), NIH/NIA R21AG083328 (SCP).

Author Contributions Statement. Xenium data was generated by ADR, KDM, SM, supervised by SCP, KRM, KM. RM preprocessed the Xenium data through SpaceRanger. CF conceptualized the spaTransfer methodological framework with suggestions from SCH. The formal analysis and benchmarking was led by CF with suggestions from BG. Figures were designed by CF. Software was developed by CF with input from other co-authors. SCP, KMaynard KMartinowich, and SCH administered and supervised the project. CF and SCH wrote the original draft of the text and all co-authors edited and approved the final manuscript.

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Competing Interests. The authors declare that they have no competing interests.

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